

What Drives Engagement on YouTube?

A Data Analysis of Influencer Metrics

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Introduction

Research question

What factors most strongly influence engagement on YouTube?

Examples of factors

- Number of views
- How long people watch
- Day of the week posted
- Number of likes and comments

Why this matters

Understanding what drives engagement helps us:

- Make better content
 - Post at the right time
 - Grow our audience faster
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Why is our topic economically relevant?

- Influencer marketing is a multi-billion-euro industry
 - Engagement influences income
 - Better allocation of marketing budgets
 - Content performance = business performance
 - Platform algorithms reward engagement
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Data Overview

What data did we analyze?

- Time period: September 2019 - November 2024 (5 years)
- Number of videos: About 300 videos
- Daily records: 234,889 observations
- Source: YouTube Analytics (official data)

What we measured

We looked at:

- Views: How many people watched
 - Likes: Approval
 - Comments: Active engagement
 - Watch time: How long people watched
 - Subscribers gained: New followers
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What is Engagement?

Simple definition

Engagement = How much people interact with your videos

$$\text{Engagement Rate} = \frac{\text{Likes} + \text{Comments}}{\text{Views}} \times 100$$

How do we measure if the engagement rate is good or bad?

- There is no universal standard for engagement across all youtube channels, so we used our own data
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Our Findings

First steps

Status of our data set: Messy, has gaps, duplicates, mixed-up data types

Example of what “messy” looks like:

date	views	likes	comments	watch_time	notes
2019-09-01	1500	45	(blank)	12500	(incomplete)
2019-09-01	1500	45	(blank)	12500	(duplicate row!)
2019-09-02	NULL	50	3	(text!)	(broken data)

What we did: Clean up this mess into usable files

Cleaning the Data

Here's what we fixed:

- **Fixed data types:** Converted text numbers to real numbers
 - So we can do math on them
- **Converted dates:** Turned text dates into proper date format
 - Extracted year, month, day of week for analysis
- **Removed incomplete rows:** Deleted rows with zero views
 - Can't calculate engagement rates without views
- **Fixed missing values:** Replaced blanks with 0

Result: Ended up with 4 clean data sets

Created Engagement Metrics

We created new columns to measure engagement:

- **Engagement Rate** = $(\text{Likes} + \text{Comments}) \div \text{Views} \times 100$
 - Example: 50 likes + 5 comments $\div$ 1000 views = **5.5%**
- **Like Rate** = $\text{Likes} \div \text{Views} \times 100$
- **Comment Rate** = $\text{Comments} \div \text{Views} \times 100$
- **Engagement Score** = $(\text{Comments} \times 2 + \text{likes} + \text{shares} \times 3) / \text{views} \times 100$

Why? With raw numbers it's difficult to compare videos. With rates it isn't.

Example:

The raw CSV has:

- **views** = 1000
- **likes** = 50
- **comments** = 5

But you can't directly answer questions like:

- “Is 50 likes good or bad for 1000 views?”
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Removed Duplicates & Aggregated

- **Removed duplicate rows:** Deleted identical records that appeared twice
- **Aggregated by video:** Collapsed 234,889 daily records into **~300 videos**
 - **Before:** Each row is one day for one video. So if a video has been up for 100 days, it had 100 rows.
 - **Now:** 1 row per video with:
 - Calculated totals: total views, total likes, total comments
 - Calculated averages: average engagement per video

- Found: first publish date, last recorded date, days active
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Average Rates

Average & Median Engagement Rate

Avg Engagement	Median Engagement
3.32%	1.89%

- We look at them to see what is ‘good’ or ‘bad’
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Like and Comment Rates

Like Rate	Comment Rate
3.12%	0.21%

Key statistics

- Average engagement: 3.32% (meaning on average, 3.32 out of 100 viewers like or comment)
- Median engagement: 1.89% (middle value - many videos have lower engagement)
- Likes are more common: About 1.76% average like rate
- Comments are rare: Only 0.09% average comment rate (comments are 15-20x less common than likes!)

What this means

Anything above **3.32%** is above our channel baseline (good); anything below is below average.

Keep in mind this benchmark is specific to our data. Factors we did not account for include:

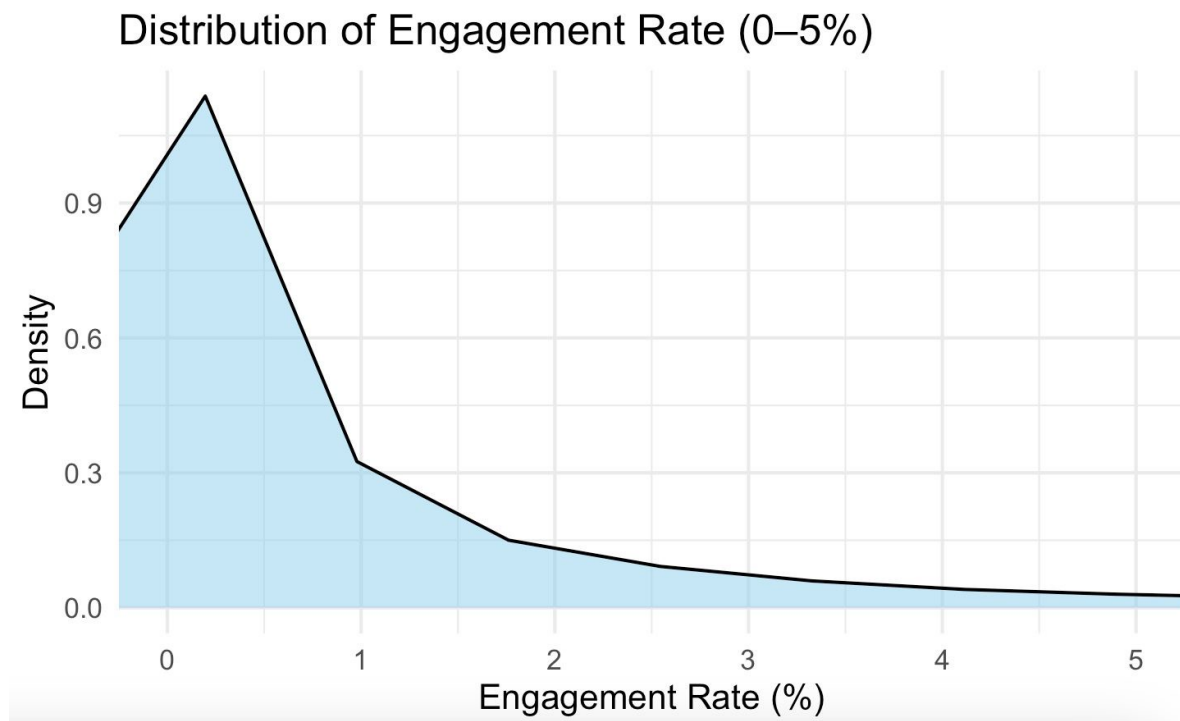
- Channel size
 - Content type
 - Audience loyalty
 - Upload frequency
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Review

Difficulties

- Data cleaning took longer than expected
 - Messy & complicated at first
 - Packages didn't load correctly (many errors)
 - Only defined engagement rate and not all additional metrics early on
 - Sharing the R project and reproducibility steps remain
 - R & data analysis tools were new
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Failed Diagram



- Used the uncleaned data
 - Wrong dates
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What Comes Next

Now we dig deeper to find what **actually drives** engagement:

- **Correlation Analysis**
 - Which metrics are connected?
 - Do longer videos get more likes?
 - Do videos posted on weekends get more engagement?
 - Does watch time predict comments?
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- **Visualizations & Graphs**

- Show trends visually
 - Line charts: Engagement over time
 - Bar charts: Performance by day of week
 - Scatter plots: Relationships between variables
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- **Key Findings & Recommendations**

- Which factors matter most for engagement
 - Actionable recommendations for better content
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Thank you so much for listening!